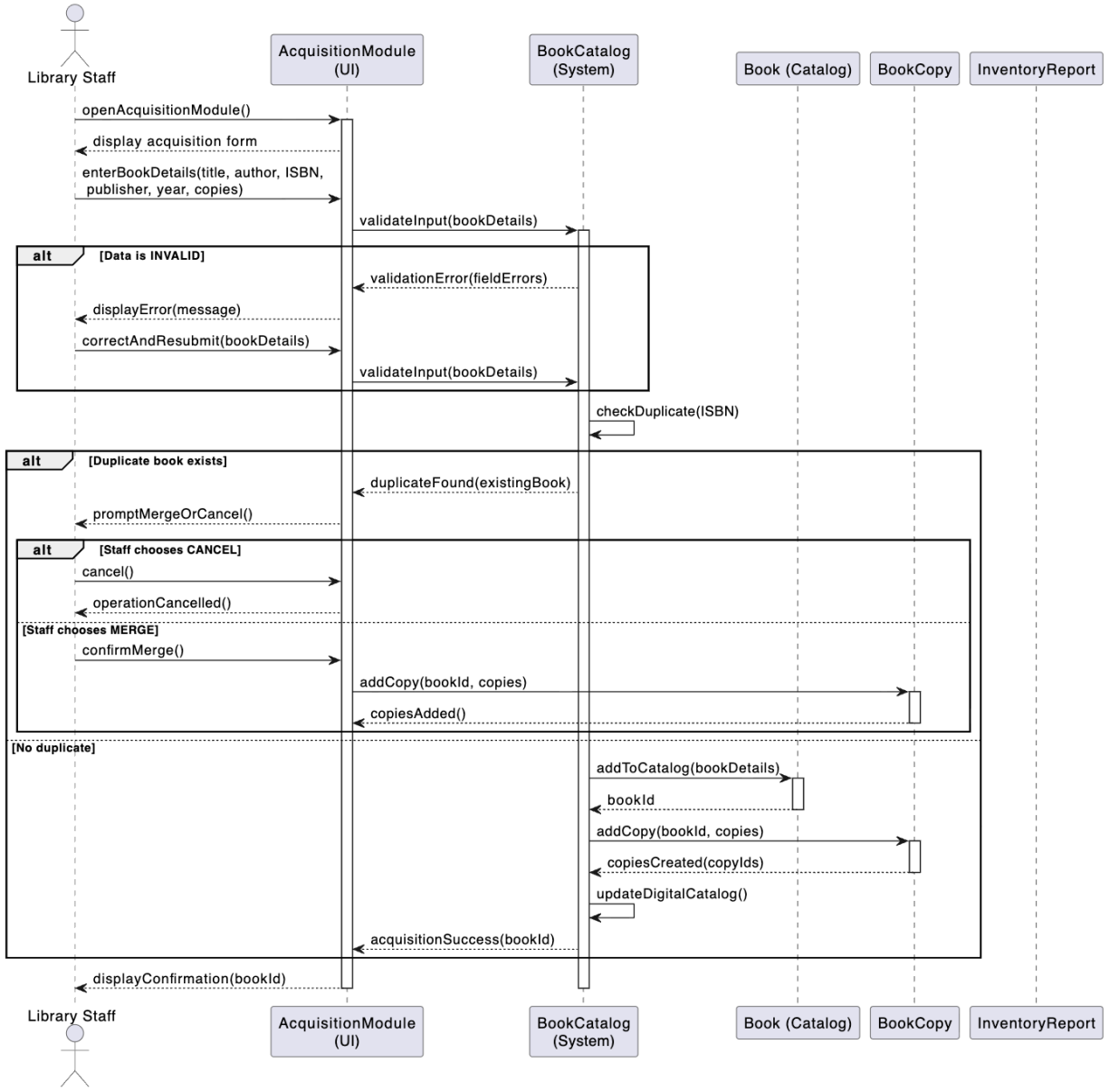
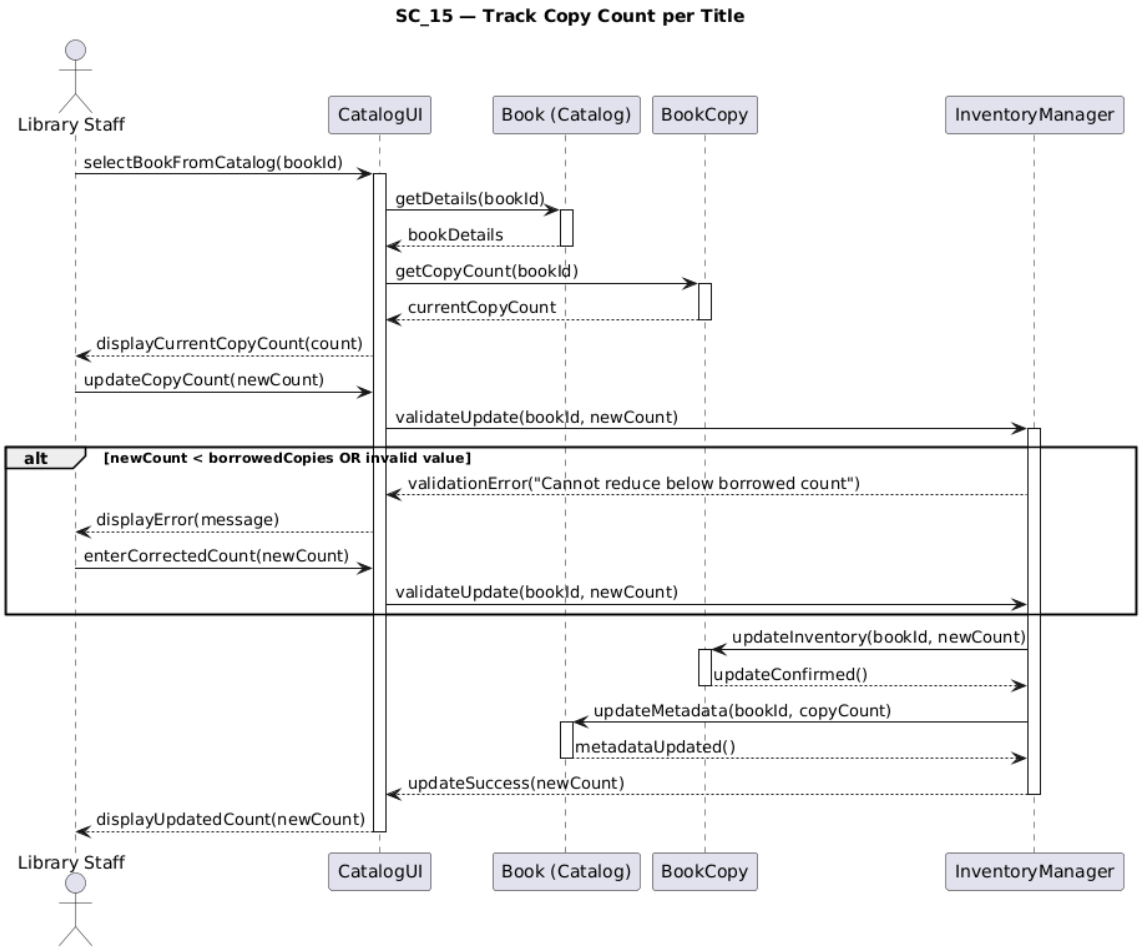
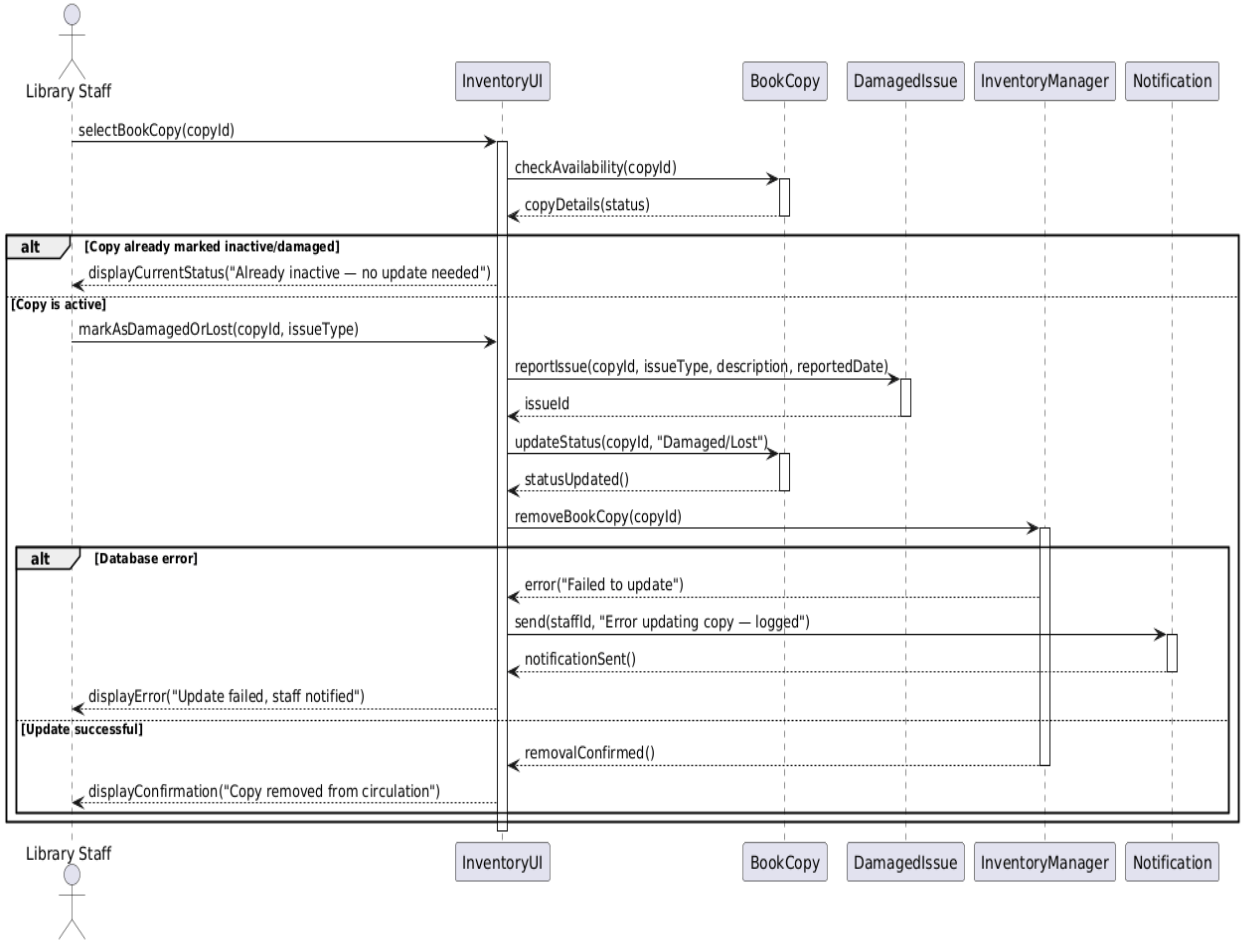
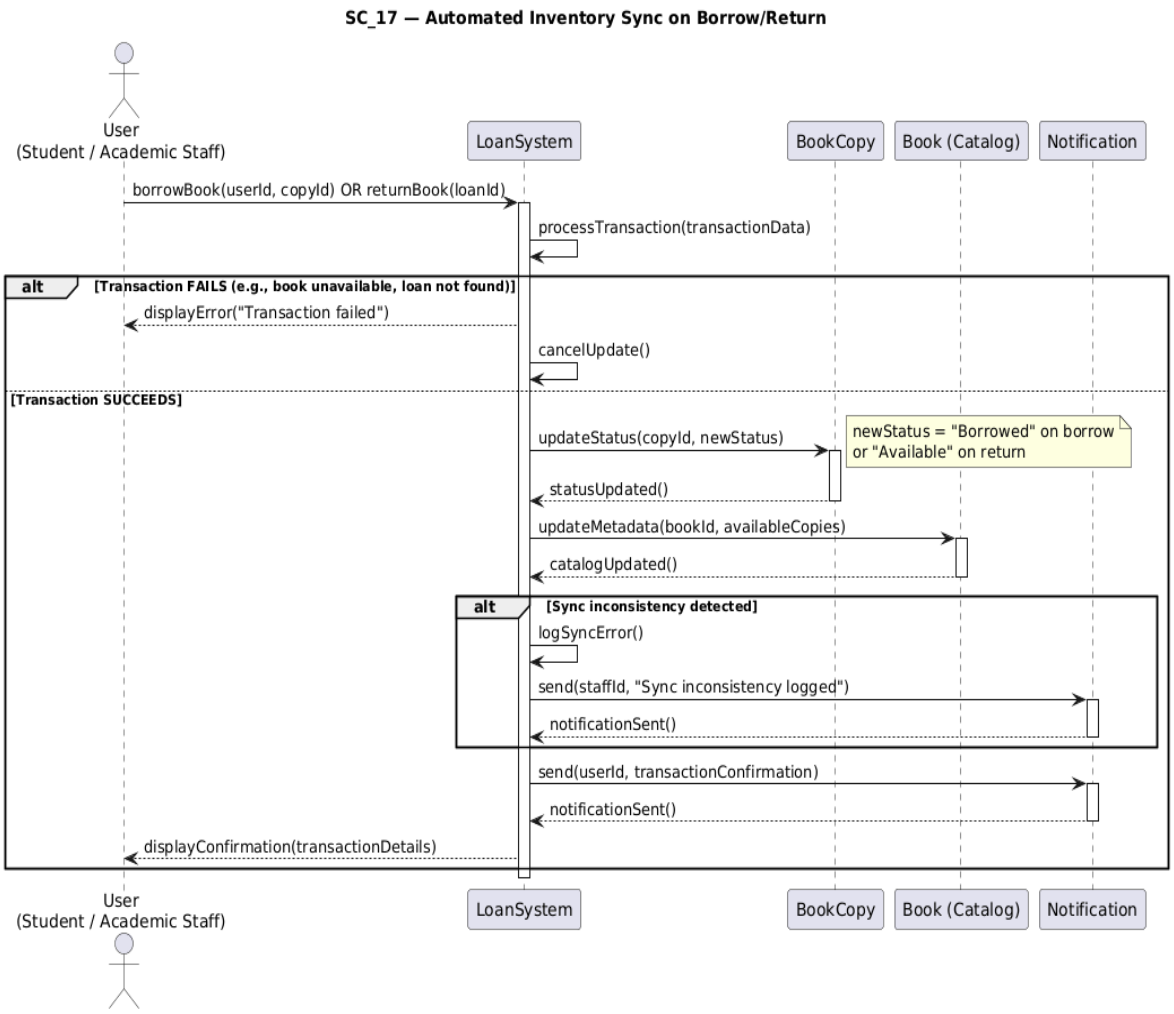
SC_14 — Record New Book Acquisition



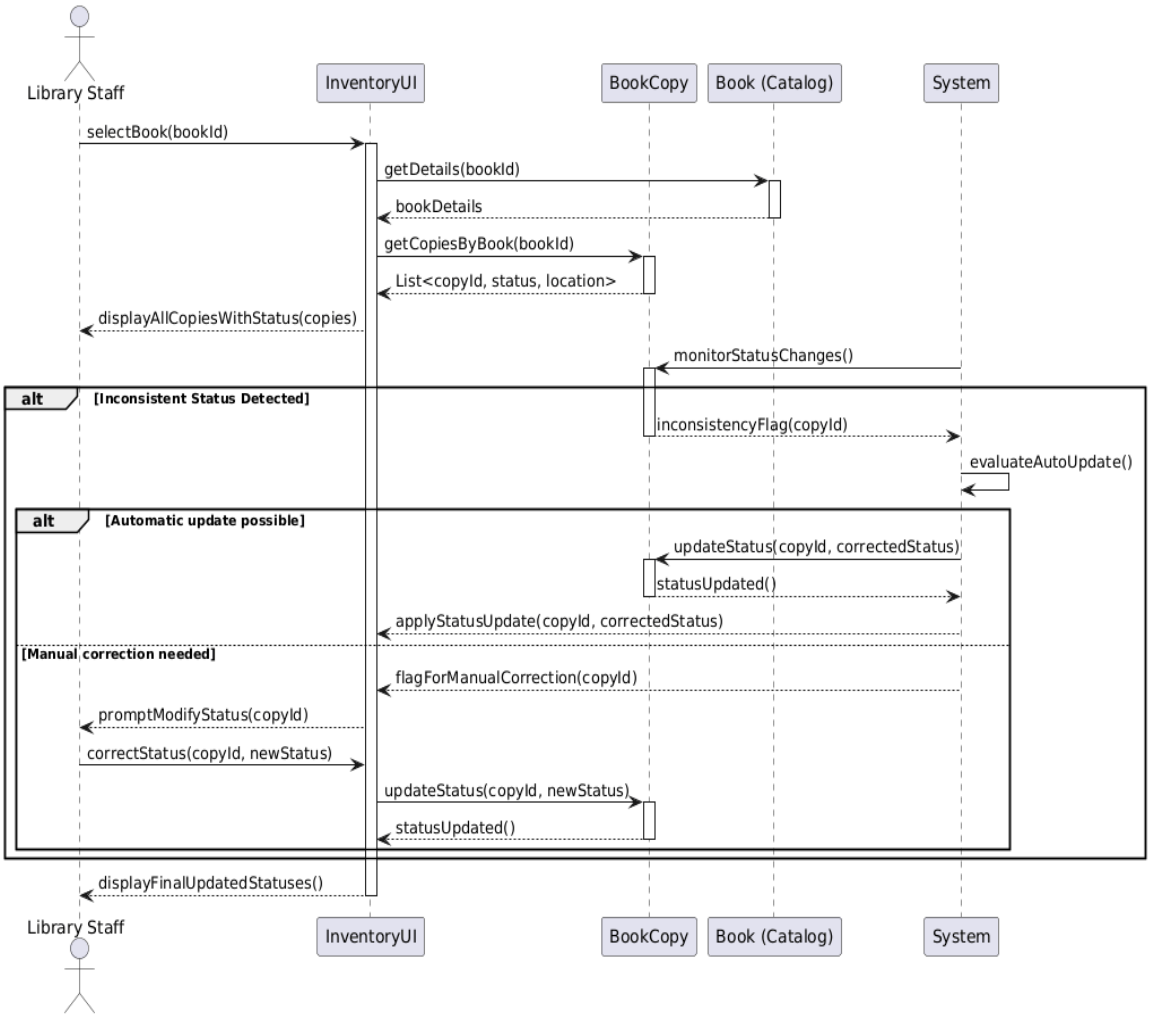

SC_15 — Track Copy Count per Title
Library Staff
CatalogUI
Book (Catalog)
BookCopy
InventoryManager
selectBookFromCatalog(bookId)
getDetails(bookId)
bookDetails
getCopyCount(bookId)
currentCopyCount
displayCurrentCopyCount(count)
updateCopyCount(newCount)
validateUpdate(bookId, newCount)
alt
[newCount < borrowedCopies OR invalid value]
validationError("Cannot reduce below borrowed count")
displayError(message)
enterCorrectedCount(newCount)
validateUpdate(bookId, newCount)
updateInventory(bookId, newCount)
updateConfirmed()
updateMetadata(bookId, copyCount)
metadataUpdated()
updateSuccess(newCount)
displayUpdatedCount(newCount)
Library Staff
CatalogUI
Book (Catalog)
BookCopy
InventoryManager

SC_16 — Manage Damaged or Lost Copies

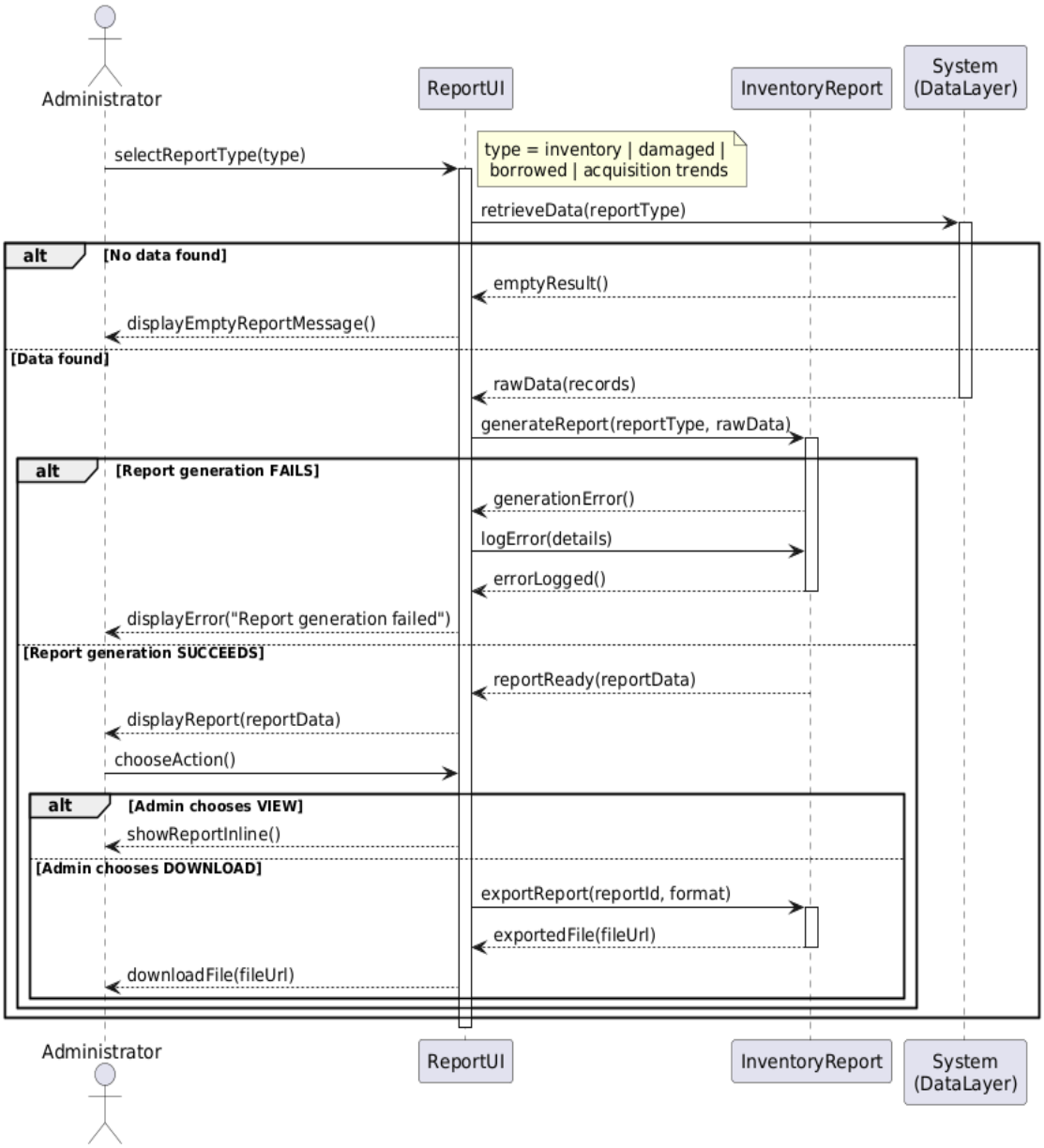
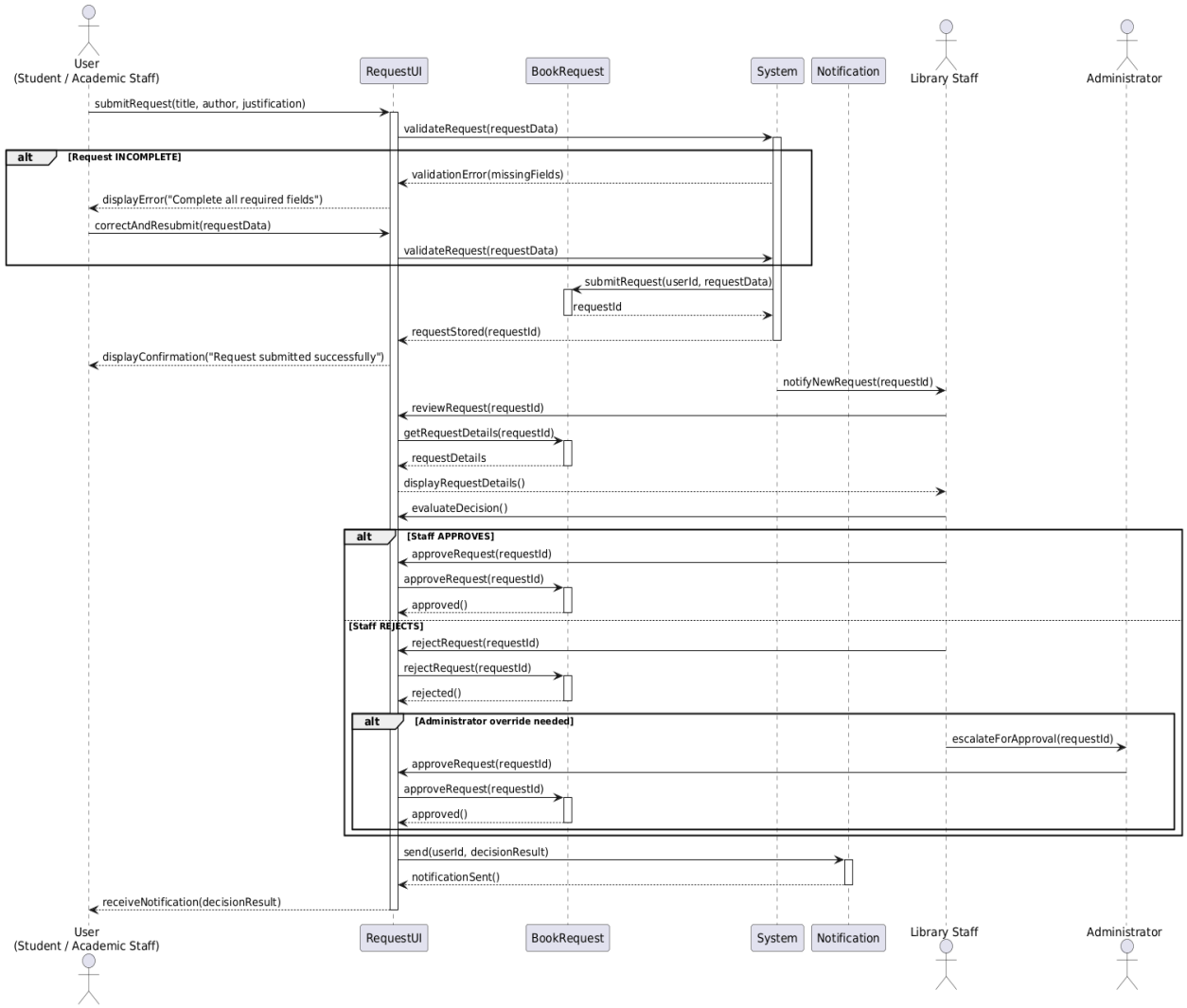



SC_17 — Automated Inventory Sync on Borrow/Return
User
(Student / Academic Staff)
LoanSystem
BookCopy
Book (Catalog)
Notification
borrowBook(userId, copyId) OR returnBook(loanId)
processTransaction(transactionData)
alt
[Transaction FAILS (e.g., book unavailable, loan not found)]
displayError("Transaction failed")
cancelUpdate()
[Transaction SUCCEEDS]
updateStatus(copyId, newStatus)
newStatus = "Borrowed" on borrow
or "Available" on return
statusUpdated()
updateMetadata(bookId, availableCopies)
catalogUpdated()
alt
[Sync inconsistency detected]
logSyncError()
send(staffId, "Sync inconsistency logged")
notificationSent()
send(userId, transactionConfirmation)
notificationSent()
displayConfirmation(transactionDetails)
User
(Student / Academic Staff)
LoanSystem
BookCopy
Book (Catalog)
Notification

SC_18 — Track Status per Physical Copy



SC_20 — Generate Inventory Reports



SC_19 — Formal Acquisition Request Process